

Random experiment: An experiment is said to be random if its results cannot be determined with certainty before hand.

Sample space: The sample space Ω is a nonempty set that contains (atleast) all possible outcomes of the random experiment.

Ex: Roll a die one time

Sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$

Ex: Roll a die until 6 show up. Record no of necessary rolls until first appearance of '6'.

$\Omega = \{1, 2, \dots, \dots\} = \mathbb{N}$

Ex: time bulb burns. $\Omega = (0, \infty)$.

If we have a defective bulb $\Omega = [0, \infty)$

Events: Subsets of a sample space.

$\mathcal{P}(\Omega)$, Power set of Ω is collection of all possible events.

Let $A \subseteq \Omega$ & $\omega \in \Omega$.

If $\omega \in A$, then we say event A occurs.

If $\omega \notin A$ then we say event A not occurs.

Ex: Experiment: Roll a die

$A = \{2, 4\}$. Suppose outcome is 6, then A did not occur.

Ex: time bulb burns. $\Omega = (0, \infty)$.

$A = [T, \infty)$ tells us light bulb burned after time T .

σ -Field: Let $\mathcal{B}$ be collection of subsets of $\mathcal{C}$.

We say $\mathcal{B}$ is a σ -field if

i) $\emptyset \in \mathcal{B}$ [$\mathcal{B}$ is nonempty]

ii) If $C \in \mathcal{B}$ then $C^c \in \mathcal{B}$ [closed under complements]

iii) If the sequence of sets $\{C_1, C_2, \dots\}$ is in $\mathcal{B}$, then $\bigcup_{i=1}^{\infty} C_i \in \mathcal{B}$ [$\mathcal{B}$ is closed under countable unions].

Examples: 1. $\mathcal{C}$ be any set $C \subseteq \mathcal{C}$. Then

$\mathcal{B} = \{C, C^c, \emptyset, \mathcal{C}\}$ is σ -field

2. $\mathcal{C}$ be any set. $\mathcal{B}$ be power set of $\mathcal{C}$, then $\mathcal{B}$ is σ -field

Probability: Let $\mathcal{C}$ be a sample space and let $\mathcal{B}$ be a σ -field. Let P be a real valued function defined on $\mathcal{B}$. The P is a probability set function if P satisfies the following

conditions: i) $P(C) \geq 0 \quad \forall \quad C \in \mathcal{B}$

ii) $P(\mathcal{C}) = 1$

iii) If $\{C_n\}$ is sequence of sets in $\mathcal{B}$ and

$C_m \cap C_n = \emptyset$ for all $m \neq n$, then

$$P\left(\bigcup_{n=1}^{\infty} C_n\right) = \sum_{n=1}^{\infty} P(C_n)$$

Example: $\Omega = \{H, T\} \quad \mathcal{F} = \{\emptyset, \{H\}, \{T\}, \Omega\}$

Define $P: \mathcal{F} \rightarrow \mathbb{R}$ by $P(\emptyset) = 0$; $P(\{H\}) = 1/5$;

$P(\{T\}) = 4/5$; $P(\Omega) = 1$. Check if P is probability function.

Thm 1 : For each event $A \in \mathcal{B}$, $P(A) = 1 - P(A^c)$.

Proof: $\mathcal{S} = A \cup A^c$ & $A \cap A^c = \emptyset$.

$$1 = P(\mathcal{S}) = P(A \cup A^c) = P(A) + P(A^c)$$

$$\therefore P(A) = \underline{\underline{1 - P(A^c)}}$$

Thm 2 $P(\emptyset) = 0$.

$$\text{Let } A = \emptyset \quad A^c = \mathcal{S} \quad \mathcal{S} = \emptyset \cup \mathcal{S}$$

$$P(\emptyset) = 1 - P(\mathcal{S}) = \underline{\underline{0}}$$

$$P(\mathcal{S}) = P(\emptyset) + P(\mathcal{S})$$

$$1 = P(\emptyset) + 1$$

Thm 3 : If A_1 & A_2 are events s.t. $A_1 \subseteq A_2$,
then $P(A_1) \leq P(A_2)$.

$$A_2 = A_1 \cup (A_1^c \cap A_2) \quad A_1 \cap (A_1^c \cap A_2) = \emptyset$$

$$\therefore P(A_2) = P(A_1) + P(A_1^c \cap A_2). \text{ Since } P(A_1^c \cap A_2) \geq 0$$

$$\Rightarrow P(A_2) \geq P(A_1).$$

Corollary : For $A \in \mathcal{B}$ $0 \leq P(A) \leq 1$

Conditional Probability

Q.1. Consider experiment of tossing 3 coins (fair). The sample space is $\Omega = \{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}$.

E be the event that at least two heads appear

F be the event that first coin show tail.

Find $P(E)$, $P(F)$ & $P(E|F)$, $P(F|E)$.

$$E = \{HHH, HHT, HTH, THH\}$$

$$F = \{TTH, THT, TTH, TTT\}.$$

$$P(E) = \frac{4}{8} = \frac{1}{2}; P(F) = \frac{1}{2}$$

Now suppose we are given that first coin is tail, i.e. F occurs, then what is probability that E occurs?

$\therefore$ New sample space is $\Omega_{\text{New}} = \{TTH, THT, TTH, TTH\}$.

$$P_{\text{New}}(E) = \frac{1}{4}$$

This kind of probability is called conditional probability of E given that F has already occurred & is denoted by $P(E|F)$ and is defined by,

$$P(E|F) = \frac{P(E \cap F)}{P(F)}, \text{ where } P(F) > 0.$$

Example: A hand of five card is to be dealt at random without replacement from an ordinary deck of 52 cards. ^{What is} the conditional probability of an all spade hand, relative to hypothesis that there are at least 4 spade in hand.

Solution: Let B be event that all 5 cards are spade.
A be event that 4 are spades & other card can be any.

$$\therefore P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(B)}{P(A)}$$

$$\text{Now } P(B) = \frac{{}^{13}C_5}{{}^{52}C_5} \quad \text{and} \quad P(A) = \frac{{}^{13}C_4 ({}^{39}C_1) + {}^{13}C_5}{{}^{52}C_5}$$

$$\therefore P(B|A) = 0.0441$$

Multiplication Rule: $P(A \cap B) = P(B|A) P(A)$.

Independent event: Sometimes it happens that the occurrence of event A does not change the probability of event B; that is $P(B|A) = P(B)$. In this case we say the events A & B are independent.

Using multiplication rule we get $P(A \cap B) = P(B) \cdot P(A)$.

Definition: Let A & B be two events. We say that A & B are independent if $P(A \cap B) = P(A) \cdot P(B)$.

Example: Let experiment be tossing 3 fair coins.

E be event that first coin is Head.

F be event that 2nd coin is Head.

i) What is probability of event F?

ii) What is probability of event F given that event E has occurred?

iii) Are E & F independent events?

Pairwise independent events: A sequence of events $\{A_i \mid i=1, 2, \dots\}$ is said to be pairwise independent events if for any A_i, A_j $P(A_i \cap A_j) = P(A_i) \cdot P(A_j)$ for all $i \neq j$.

Mutually independent events: A sequence of events $\{A_i \mid i=1, 2, \dots\}$ is said to be mutually independent iff for any finite subcollection $A_1, \dots, A_k$, $P(A_1 \cap \dots \cap A_k) = P(A_1) \cdot \dots \cdot P(A_k)$ for all k .

Example: Let $\Omega = \{1, 2, 3, 4\}$, $A = \{1, 2\}$, $B = \{1, 3\}$, $C = \{1, 4\}$,
Let $P(\{w\}) = \frac{1}{4}$ where $w = 1, 2, 3, 4$.

$$A \cap B = \{1\} = A \cap C = B \cap C = A \cap B \cap C.$$

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{2}; P(C) = \frac{1}{2}$$

$$P(A \cap B) = \frac{1}{4}; P(A \cap C) = \frac{1}{4}; P(B \cap C) = \frac{1}{4}; P(A \cap B \cap C) = \frac{1}{4}$$

$$P(A \cap B \cap C) = \frac{1}{4} \neq P(A)P(B)P(C) = \frac{1}{8}$$

$\therefore A, B, C$ are pairwise independent but not ^{mutually} ~~pairwise~~ independent.

Definition! Baye's Rule!

For any event $B \in \mathcal{F}$ with $P(B) > 0$ and $\{A_i\}$ be partition with $P(A_i) > 0$, then $P(A_i|B) = \frac{P(A_i) P(B|A_i)}{\sum_j P(A_j) P(B|A_j)}$

Example: Consider a city with 45% females and 55% males. Suppose 70% of males & 10% of females smoke. Find the probability that a smoker is male.

Solution: Let S be ^{the} event that person is a smoker.
 M be ^{the} event that person is a male.
 F be the even that female.

$$P(M) = \frac{55}{100} ; P(F) = \frac{45}{100} ; P(S|M) = \frac{70}{100} ; P(S|F) = \frac{10}{100}$$

$$\begin{aligned} \text{To find } P(M|S) &= \frac{P(M) P(S|M)}{P(M) P(S|M) + P(F) P(S|F)} \\ &= \frac{\frac{55}{100} \times \frac{70}{100}}{\frac{55}{100} \times \frac{70}{100} + \frac{45}{100} \times \frac{10}{100}} = \frac{55 \times 70}{(55 \times 70) + (45 \times 10)} \\ &= \underline{\underline{0.895}} \end{aligned}$$

Homework: A bag contains 75 fair coins & 25 unfair coins with $P(H) = 0.9$ & $P(T) = 0.1$. A coin is picked at random & tossed n -times & each one of the n -tosses were heads.

- i) what is the probability that the picked coin is an unfair coin?
- ii) Find the least value of n that gives probability that the picked coin is unfair to be at least 80%.

Bayes Rule problem on unfair coin.

Ans: i) Define events:

~~Ans~~ ii) U : Picked coin is unfair

F : Picked coin is fair

H^n : all n tosses came head.

Total coins given is 100.

$$\therefore P(U) = \frac{25}{100} = 0.25$$

$$P(F) = \frac{75}{100} = 0.75$$

Likelihoods:

If coin is fair

$$P(H^n|F) = \left(\frac{1}{2}\right)^n$$

$$P(H^n|U) = (0.9)^n$$

We need to find $P(U|H^n) = \frac{P(U) \cdot P(H^n|U)}{P(H^n|U)P(U) + P(H^n|F)P(F)}$

$$= \frac{0.25}{0.25 + 0.75 \left(\frac{5}{9}\right)^n}$$

$$\text{ii)} \quad \frac{0.25}{0.25 + 0.75 \left(\frac{5}{9}\right)^n} > 0.8$$

$$\Rightarrow 0.25 > 0.8 \left[0.25 + 0.75 \times \left(\frac{5}{9}\right)^n \right]$$

$$= 0.2 + 0.6 \left(\frac{5}{9}\right)^n$$

$$\Rightarrow 0.05 \geq 0.6 \left(\frac{5}{9}\right)^n$$

$$\Rightarrow \frac{0.05}{0.6} \geq \left(\frac{5}{9}\right)^n$$

$$\Rightarrow 0.0833 \geq \left(\frac{5}{9}\right)^n$$

$$\text{ie: } \log\left(\frac{0.05}{0.6}\right) \geq n \log\left(\frac{5}{9}\right)$$

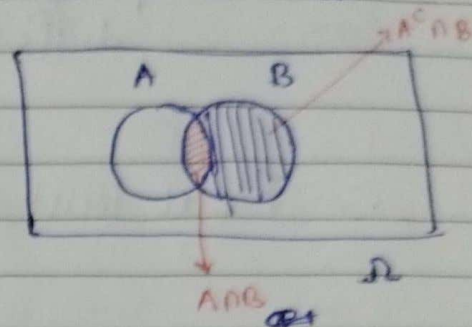
$$n \geq \underline{\underline{4.2276}}$$

$$\Rightarrow \log 4.2275 \geq n$$

$$\therefore n = 5$$

Question : If A & B are independent events. Show that

- i) A^c, B is independent
- ii) A, B^c is independent
- iii) A^c, B^c is independent



Proof: i) Note that $B = (A \cap B) \cup (A^c \cap B)$

$$\text{also } (A \cap B) \cap (A^c \cap B) = \emptyset$$

$$\therefore P(B) = P(A \cap B) + P(A^c \cap B)$$

$$\begin{aligned} \therefore P(A^c \cap B) &= P(B) - P(A \cap B) \\ &= P(B) - P(A) \cdot P(B) \\ &= P(B) [1 - P(A)] \\ &= P(B) P(A^c) \end{aligned}$$

$$\begin{aligned} &(A^c \cap B) \cup (A \cap B) \\ &= [(A^c \cap B) \cup A] \cap [(A^c \cap B) \cup B] \\ &= [(A^c \cup A) \cap (B \cup A)] \cap [(A^c \cup B) \cap (B \cup A)] \\ &= (B \cup A) \cap [(A^c \cap B) \cup (B \cap B)] \end{aligned}$$

$\therefore A^c, B$ are independent.

Question: Assuming that a year has 365 days, what is the probability that in a room with 5 people, ~~there are two~~ at least of them with same birthday?

Solution: E be event that at least two people share same birthday.

E^c be event that all 5 people have different birthday.

$$\therefore P(E) = 1 - P(E^c)$$

$$P(E^c) = \frac{365}{365} \times \frac{364}{365} \times \frac{363}{365} \times \frac{362}{365} \times \frac{361}{365}$$

$$\therefore P(E) = 1 - P(E^c) = 1 - 0.9729 = \underline{\underline{0.0271}}$$

Random Variable:

Sample space is difficult to describe if elements are not numbers. Now we formulate results to represent $c \in \mathcal{C}$ as numbers.

Definition: Let $\mathcal{C}$ be a sample space for a random experiment. A function $X: \mathcal{C} \rightarrow \mathbb{R}$ is called random variable.

Range $(X) = \mathcal{A}$ which we assume to be

- i) countable set ii) intervals in $\mathbb{R}$

If $\mathcal{A}$ is countable, X is called discrete random variable.

If $\mathcal{A}$ is interval, X is called continuous random variable.

If X is discrete random variable with range $\mathcal{A} = \{d_1, \dots, d_m\}$, then there are ' m events of interest' ie

$$\{c \in \mathcal{C} \mid X(c) = d_i\} \quad \text{for } i = 1, 2, \dots, m.$$

$\therefore$ For this random variable, the σ -field

on $\mathcal{A}$ is one generated by $\{\{d_1\}, \{d_2\}, \dots, \{d_m\}\}$ denoted by $\mathcal{F}$.

Probability induced by X : For $B \in \mathcal{F}$, define

$$P_x(B) = P(\{c \in \mathcal{C} \mid X(c) \in B\})$$

$$= \sum_{d_i \in B} P(\{c \in \mathcal{C} \mid X(c) = d_i\})$$

[$\because B$ is subset of $\mathcal{A}$]

$$= P_x(\{d_i\})$$

$\therefore P_x$ is completely determined by the fn,

$$p_x(d_i) = P_x(\{d_i\}) \quad i = 1, 2, \dots, m.$$

This fn is called the probability mass function of X [pmf].

Analogue:

Sample space	$\mathcal{C}$	$\mathcal{A} = X(\mathcal{C})$
σ -Field	$\mathcal{P}(\mathcal{C})$	generated by $\{\{d_1\}, \dots, \{d_m\}\}$
Probability	P	PMF

Example: Let X be the sum of the upfaces on a roll of a pair of fair 6-sided dice, each with the numbers 1 to 6.

Sample space $\mathcal{C} = \{ (i,j) \mid 1 \leq i,j \leq 6 \}$

$\therefore X : \mathcal{C} \rightarrow \mathbb{R}$ is $X((i,j)) = i+j$.

$\therefore \mathcal{D} = \{ 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 \}$.

We need to determine $p_x(d_i)$ for $d_i \in \mathcal{D}$.

i) $p_x(2) = P(\{c \in \mathcal{C} \mid X(c) = 2\})$.

Since $(1,1)$ is the only element s.t

$$X(1,1) = 2, \quad p_x(2) = \frac{1}{36}.$$

ii) $p_x(7) = P(\{c \in \mathcal{C} \mid X(c) = 7\})$.

We know, $\{ (1,6), (6,1), (2,5), (5,2), (3,4), (4,3) \}$ are elements whose sum is 7.

$$\therefore p_x(7) = \frac{6}{36}.$$

Note that σ -field for the probability space $\mathcal{C}$ would consist 2^{36} subsets. But for random variable X we need only 11 simple events of interest.

Suppose $B_1 = \{ 7, 11 \}$, then

$$P_x(B_1) = p_x(7) + p_x(11) = \frac{6}{36} + \frac{2}{36} = \underline{\underline{\frac{8}{36}}}.$$

ii) Continuous random variable.

Continuous random variable are often measurements not mass.

For example, we are not interested in the "probability that a person weighs 60kg", instead we are interested in "probability that person weighs over 60kg".
 $\therefore$ Events of interest are intervals.

Definition:

For any interval $(a, b) \in \mathcal{A}$, the induced probability distribution of X is defined as

$$\begin{aligned} P_x [(a, b)] &= P(\{ \omega \in \Omega \mid a < X(\omega) < b \}) \\ &= \int_a^b f_x(x) dx \end{aligned}$$

f_x is called probability density function (pdf) of X .

Example: Choose a real number at random from interval $(0, 1)$.

- Sample space $\Omega = (0, 1)$
- Define $X: \Omega \rightarrow \mathbb{R}$ by X be

Let X :

Here sample space of X is $\Theta = (0, 1)$!

$$P_X([1, 0]) = 1 = 1 - 0$$

$$\therefore P_X((a, b)) = b - a \quad \text{where } 0 < a < b < 1.$$

$$P: \mathcal{F} \rightarrow \mathbb{R}$$

$$X: \Omega \rightarrow \mathbb{R}$$

$$X(\Omega) = \Theta = (0, 1).$$

$$P_X([a, b]) = P(\{c \in (0, 1) \mid a < X(c) < b\})$$

$$P_X([0, 1]) = 1 = \int_0^1 f_X(x) dx$$

$$\Rightarrow f_X(x) = 1.$$

$$\therefore f_x(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Question: Find $P \left[\left\{ x < \frac{1}{8} \right\} \cup \left\{ x > \frac{7}{8} \right\} \right]$

$$= P \left[\left\{ C \in (0,1) \mid 0 < C < \frac{1}{8} \text{ or } \frac{7}{8} < C < 1 \right\} \right]$$

$$= \int_0^{1/8} f_x(x) dx + \int_{7/8}^1 f_x(x) dx$$

$$= \frac{1}{8} + 1 - \frac{7}{8} = \frac{1+8-7}{8} = \frac{2}{8} = \underline{\underline{\frac{1}{4}}}$$

Cumulative Distribution Function (CDF).

Let X be r.v. Then its C.D.F is defined by $F_x(x)$, where

$$F_x(x) = P_x \left((-\infty, x] \right) = P \left(\{ C \in \Omega \mid X(C) \leq x \} \right).$$

Example:

1. Suppose we roll a fair die. Let X be upface of the roll.

$$\therefore \mathcal{A} = \{1, 2, 3, 4, 5, 6\}.$$

$$p_x(i) = \frac{1}{6}.$$

$$\text{Now } F_x(x) = p_x((-\infty, x])$$

$$= p(\{c \in \mathcal{C} \mid x(c) \leq x\})$$

i) If $x < 1$

$$F_x(x) = p_x((-\infty, x])$$

$$= p(\{c \in \mathcal{C} \mid x(c) \leq 1\})$$

$$= 0$$

$$\text{If } 1 \leq x < 2 \quad F_x(x) = \frac{1}{6}$$

Expectation of a random variable

Let X be a random variable.

i) If X is a continuous r.v with pdf $f_x(x)$ and $\int_{-\infty}^{\infty} |x| f(x) dx < \infty$,

then Expectation of X , is

$$E(x) = \int_{-\infty}^{\infty} x f(x) dx. \quad \text{is}$$

ii) If X is a discrete random variable with pmf $p_x(x)$ and $\sum_x |x| p_x(x) < \infty$,

then the Expectation of X is

$$E(x) = \sum_x x p_x(x)$$

Note: Expectation of X is also called mean & denoted by $\mu = E(X)$.

Examples:

1. Let the random variable X of the discrete type have the pmf given below

x	1	2	3	4
$p(x)$	$\frac{4}{10}$	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{2}{10}$

and $p(x)=0$ elsewhere

$$\therefore E(x) = \sum x p_x(x)$$

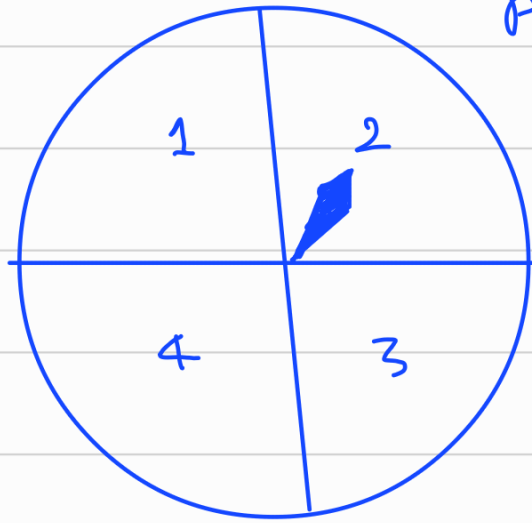
$$= 1 \times \frac{4}{10} + 2 \times \frac{1}{10} + 3 \times \frac{3}{10} + 4 \times \frac{2}{10}$$

$$= \frac{23}{10} = 2.3$$

2. Let the continuous r.v X have the pdf
- $$f(x) = 4x^3 \quad 0 < x < 1$$
- $$= 0 \quad \text{elsewhere.}$$

$$E(x) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^1 4x^4 dx = 4 \left[\frac{x^5}{5} \right]_0^1$$
$$= \frac{4}{5}$$

How did word 'Expectation' origin?



$$P(\text{arrow stops at 1}) = 0.2$$

$$P(2) = 0.30$$

$$P(3) = 0.35$$

$$P(4) = 0.15$$

1. Deposit 50 Rs

2. Spin the wheel

3. Arrow at 1, you win 20 Rs

2, —||— 30 Rs

3, —||— 40 Rs

4, —||— 120 Rs.

4. Net gain : at 1, $20 - 50 = -30$ Rs

2, $30 - 50 = -20$ Rs

3, $40 - 50 = -10$ Rs

4, $120 - 50 = 70$ Rs.

5. $P(-30) = 0.20$; $P(-20) = 0.30$

$$P(-10) = 0.35 \quad ; \quad P(70) = 0.15.$$

$$\begin{aligned} \therefore E(G) &= (-30) \times 0.2 + (-2)(0.3) + (-10)(0.35) \\ &\quad + (70)(0.15) \\ &= 10 [-0.50] = \underline{\underline{-5.0}} \end{aligned}$$

Question: A number X is selected from $\{1, 2, \dots, 2n+1\}$. Find $E(X)$.

Let X be r.v of selecting a number from $\{1, 2, \dots, 2n+1\}$

Solution : $X : \Omega \rightarrow \mathbb{R}$, $\Omega = \{1, 2, \dots, 2n+1\}$
where $X(d) = d$.

Now $\Omega = \{1, 2, \dots, 2n+1\}$

$$\begin{aligned} p_x(1) &= P(\{c \in \Omega \mid X(c) = 1\}) \\ &= P(\{1\}) = \frac{1}{2n+1} \end{aligned}$$

In general $p_x(d) = \frac{1}{2n+1}$

$$\begin{aligned} \therefore E(X) &= 1 \times \frac{1}{2n+1} + 2 \times \frac{1}{2n+1} + \dots + (2n+1) \frac{1}{2n+1} \\ &= \frac{1}{2n+1} [1 + 2 + \dots + 2n+1] \end{aligned}$$

$$= \frac{1}{2n-1} \left[\frac{(2n-1)(2n-1+1)}{2} \right]$$

$$= \frac{2n}{2} = \underline{\underline{n}}$$

Properties of Expectations:

1. Let X be constant random variable, ie $X(c) = k \quad \forall c \in \Omega$.
Find expectation of X .

Sol: Since X is constant $\mathcal{R} = \{k\}$

$$E(X) = \sum x p_x(x) = k p_x(k)$$

$$\begin{aligned} p_x(k) &= P(\{c \in \Omega \mid X(c) = k\}) \\ &= P(\Omega) = 1 \end{aligned}$$

$$\boxed{\therefore E(X) = k}$$

2. If X_1 & X_2 are two r.v $\in \alpha, \beta \in \mathbb{R}$
then $E(\alpha X_1 + \beta X_2) = \alpha E(X_1) + \beta E(X_2)$.

Variance: Let X be r.v with finite mean μ & st $E[(X-\mu)^2]$ is finite. Then variance of X is defined to be $E[(X-\mu)^2] = \sigma^2$

$$\begin{aligned}\text{Note : 1. } E[(X-\mu)^2] &= E(X^2 + \mu^2 - 2\mu X) \\ &= E(X^2) + E(\mu^2) - 2\mu E(X) \\ &= E(X^2) + E(\mu^2) - 2\mu^2 \\ &= E(X^2) + \mu^2 E(1) - 2\mu^2 \\ &= E(X^2) - \mu^2.\end{aligned}$$

2. σ is called standard deviation.

Question: A number X is selected from $\{1, 2, \dots, 2n+1\}$. Find σ^2 .
Let X be r.v of selecting a number from $\{1, 2, \dots, 2n+1\}$

$$\text{Sol: } E[(X-\mu)^2] = E(X^2) - \mu^2.$$

$$\mu = n \quad \therefore \mu^2 = n^2$$

$$\begin{aligned}X^2: \Omega &\rightarrow \mathbb{R} \text{ defined by} \\ X^2(\omega) &= (X(\omega))^2\end{aligned}$$

$$\therefore A' = \{1, 2^2, \dots, (2n-1)^2\}$$

$$E(x^2) = \sum x^2 p_x(x)$$

$$= \left(1 \times \frac{1}{2n-1}\right) + 2^2 \times \frac{1}{2n-1} + \dots + (2n-1)^2 \times \frac{1}{(2n-1)}$$

$$= \frac{1}{2n-1} \times [1 + 2^2 + \dots + (2n-1)^2]$$

$$= \frac{1}{2n-1} \times \frac{(2n-1) \times (2n) (2(2n-1) + 1)}{6}$$

$$= \frac{1}{\cancel{(2n-1)}} \times \frac{\cancel{(2n-1)} (2n) (4n-1)}{6}$$

$$= \frac{4n^2 - n}{3}$$

$$\therefore G^2 = E(x^2) - \mu^2 = \frac{4n^2 - n}{3} - n^2$$

$$= \frac{4n^2 - n - 3n^2}{3} = \underline{\underline{\frac{n^2 - n}{3}}}$$

$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$
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Some Special distribution:

Bernoulli experiment: A random experiment in which ^{result} ~~outcome~~ has only two possible outcomes, called "success" & "failure", where ~~the probability of success say~~ p , ~~remain the same from trial to trial.~~ is B.E.

Success or failure can be death or life, female or male, nondefective or defective, head or tail.

Bernoulli trial: A sequence of Bernoulli trial occurs when a Bernoulli expt is performed several independent times. ~~see that~~

In such ^{expt} ~~trial~~, the probability of success say p , remains same from trial to trial.

Random variable of Bernoulli's experiment:

Let X be r.v associated with Bernoulli's trial defined as $X(\text{success}) = 1$ & $X(\text{failure}) = 0$.

$$p(0) = P(\{\omega \in \Omega \mid X(\omega) = 0\}) \quad \text{1st term}$$

~~$= P$~~

The pmf of a Bernoulli distribution is

$$p(x) = p^x (1-p)^{1-x} \quad x = 0, 1$$

The expectation $\mu = E(x) = 0 \cdot p(0) + 1 \cdot p(1)$

$$= 0(1-p) + 1(p)$$
$$= \underline{\underline{p}}$$

the variance of X is $\sigma^2 = E(X^2) - \mu^2 = E(X^2) - p^2$
 where $E(X^2) = 0$

$$= \sum (x^2) p_x(x) = 1^2 p + 0(1-p) = p^2$$

$$= p - p^2 = \underline{\underline{p(1-p)}}$$

Question: A machine produces items, & each item is defective with probability 0.1 & non-defective with probability 0.9.

- Defined a Bernoulli's r.v X for item is defective.
- Find $P(X=1)$; $P(X=0)$ & $\text{Var}(X)$.
 pmf expectation

Solution: i) Since we are interested in if item is defective
 $p = 0.1$ i.e. probability that item is defective is 0.1
 & $1 - p = 0.9$ is non-defect.

∴ Bernoulli's r.v is defined as,

$$X \text{ (defective item)} = 1$$

$$X \text{ (non defective item)} = 0$$

ii) pmf is given by $p(x) = (0.1)^x (0.9)^{1-x}$ $x=0,1$

iii) $E(X) = p = 0.1$

iv) Variance $= \sigma^2 = p(1-p) = 0.1 \times 0.9 = \underline{\underline{0.09}}$

Question: A student gives the answer to a true/false question. Let $X = \begin{cases} 1 & \text{if answer is correct} \\ 0 & \text{if answer is wrong} \end{cases}$. Define Bernoulli's r.v. X . Find pmf, expectation & Variance of X .

Sol: Note that experiment is Bernoulli's

$\therefore$ i) Random variable is defined by

$$X (\text{correct answer}) = 1$$

$$X (\text{wrong answer}) = 0.$$

ii) Here the probability that answer is correct is $p = 0.5$ & the probability that answer is wrong is $1-p = 0.5$

$$\therefore \text{pmf is } p(x) = (0.5)^x (0.5)^{1-x} \quad x=0,1.$$

$$\text{ii) } E(x) = 0.5$$

$$\text{iv) Variance } \sigma^2 = 0.5 \times (1-0.5) = 0.5 \times 0.5 = 0.25.$$

In a sequence of n independent Bernoulli trials, where the probability of success remains constant, let X_i denote the Bernoulli random variable associated with the i th trial. An observed sequence of n Bernoulli trials is then an n -tuple of zeros and ones. In such a sequence of Bernoulli trials, we are often interested in the total number of successes and not in the order of their occurrence. If we let the random variable X equal the number of observed successes in n Bernoulli trials, the possible values of X are $0, 1, 2, \dots, n$. If x successes occur, where $x = 0, 1, 2, \dots, n$, then $n - x$ failures occur. The number of ways of selecting the x positions for the x successes in the n trials is

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}.$$

Since the trials are independent and the probabilities of success and failure on each trial are, respectively, p and $1 - p$, the probability of each of these ways is $p^x(1 - p)^{n-x}$. Thus the pmf of X , say $p(x)$, is the sum of the probabilities of these $\binom{n}{x}$ mutually exclusive events; that is,

$$p(x) = \begin{cases} \binom{n}{x} p^x (1 - p)^{n-x} & x = 0, 1, 2, \dots, n \\ 0 & \text{elsewhere.} \end{cases} \quad (3.1.2)$$

It is clear that $p(x) \geq 0$. To verify that $p(x)$ sums to 1 over its range, recall the binomial series, expression (1.3.7) of Chapter 1, which is:

$$(a + b)^n = \sum_{x=0}^n \binom{n}{x} b^x a^{n-x},$$

for n a positive integer. Thus,

$$\begin{aligned} \sum_x p(x) &= \sum_{x=0}^n \binom{n}{x} p^x (1 - p)^{n-x} \\ &= [(1 - p) + p]^n = 1. \end{aligned}$$

Therefore, $p(x)$ satisfies the conditions of being a pmf of a random variable X of the discrete type. A random variable X that has a pmf of the form of $p(x)$ is said to have a **binomial distribution**, and any such $p(x)$ is called a **binomial pmf**. A binomial distribution is denoted by the symbol $b(n, p)$. The constants n and p are called the **parameters** of the binomial distribution.

Example 3.1.1 (Computation of Binomial Probabilities). Suppose we roll a fair six-sided die 3 times. What is the probability of getting exactly 2 sixes? For our notation, let X be the number of sixes obtained in the 3 rolls. Then X has a binomial distribution with $n = 3$ and $p = 1/6$. Hence,

$$P(X = 2) = p(2) = \binom{3}{2} \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^1 = 0.06944.$$